

Sacha Arseneault

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EDUCATION

University of Ottawa

BASc in Software Engineering, Co-op

Ottawa, ON

GPA: 3.9/4.0 | Expected: Dec 2026

EXPERIENCE

Microsoft AI

Applied AI/ML Intern — Copilot

Mountain View, CA

June 2026 - Aug 2026

- Modernized Copilot's agent stack serving **100M+ monthly users**—across harness engineering, response quality, and inference—driving its transition toward longer-running agentic workflows.
- Increased first-turn **KV cache hit rates** on low-traffic GPT models from 7% to 90% by introducing dynamic cross-conversation GPU routing, saving up to \$1.4M annually in inference costs.
- Architected an end-to-end evaluation platform for autonomous coding-agent fleets, profiling 1,000+ PRs/week across Microsoft products serving **over 500M monthly users**.
- Influenced the “autopilot” feature roadmap by demoing *self-directed continuity* to VP leadership, enabling always-running agents to pursue missions autonomously with **zero user intervention**.

Tesla

Software Engineer Intern — Robotaxi

Palo Alto, CA

May 2025 - Aug 2025

- Established **early reliability guardrails** for Tesla's robotaxi platform—across automation testing, observability, and rider feedback pipelines—maximizing readiness for its official release in June 2025.
- Configured post-merge canaries validating **the complete rider journey**—from ride hailing to passenger drop-off—by using mobile emulators and simulated car fleets, flagging deployment issues **immediately** without impacting live rides.
- Built an integration test suite validating telemetry emitted from 6 different sources during **real robotaxi rides**, reducing full-system verification from 15 to 2 minutes and saving 150 hrs/year of manual testing work per engineer. (Go, Docker, Kafka)
- Implemented fault-tolerant telemetry parsing that recovers 100% of valid fields in legacy payloads, preserving accountability and observability for robotaxis running on older Model Y firmware.
- Conducted **live QA across 200+ physical rides**, identifying infotainment and rider experience issues and coordinating resolution with cross-functional teams ahead of Early Access.

Shopify

Software Engineer Intern — Stores

Ottawa, ON

Jan 2025 - Apr 2025

- Strengthened Shopify's store-creation pipeline handling **2M+ annual store creations**—with data consistency safeguards and automated recovery—preventing failed or inconsistent creations from disrupting merchant launches.
- Protected 20,000+ new stores annually from customer-facing data inconsistencies by resolving a **1-in-100 race condition** that created corrupted database records during merchant onboarding.
- Eliminated the need for on-call engineers to remediate backend store-creation failures by **automating recovery across all failure states**—including timeouts and platform outages—resolving 100% of stuck requests. (Ruby, Rails, GraphQL, MySQL)

Motorola | Software Engineer Intern — Emergency Call Handling

Gatineau, QC | Sep 2024 - Dec 2024

Ciena | Software Engineer Intern — Developer Tools & Experience

Ottawa, ON | Jan 2024 - Apr 2024

LEADERSHIP

IEEE uOttawa - Webmaster

2022 - 2026

- Led a team of 10 developers in building 4 open-source web-apps driving 5,000+ visits and 40,000+ impressions per month, serving a community of over **30,000 engineering students** across Eastern Canada.
- Hosted 2 workshops (Git/React) and 3 review sessions (Data Structures and Algorithms) with 75+ attendees each.

PROJECTS

Concurrency Debugger for Go | Golang, Websocket, GinkGo

2026

- A cross-platform Go debugger that visualizes goroutines, channels, and synchronization through low-level Linux/macOS backends, exposing execution control and runtime inspection to IDEs and coding agents via DAP.

Eco Rewards App | Remix, TypeScript, Express.js, PostgreSQL, ShadCn, Auth0

Waterloo, ON | 2024

- A full-stack points-tracking app rewarding users for sustainable day-to-day actions, with AI receipt parsing.
- Won 1st Place** out of 842 participants in a top sponsor track at **Hack the North**, Canada's largest hackathon.

TECHNICAL SKILLS

Languages: Go, Python, TypeScript, Java, SQL, Ruby

Frameworks/Libraries: Express, Next, React, Redux, PostgreSQL, Rails, Supabase, FastAPI, Scikit-learn, Pandas, Torch

Tools: Git, Docker, Linux, Bash, Prometheus, Grafana, OpenTelemetry, Kafka, Redis, GraphQL, gRPC, Protobuf